

Name:
Z number:

Total Points: 20

COP 3410-001: Data Strc Algrthm Anlys Python

Florida Atlantic University

Topic: Algorithm Analysis

WORKSHEET

NOTE:

- Each question carries 1 point.
- All solutions must be answered in the space labeled “put your solutions here.”
- submit the completed worksheet in the form of a PDF.
- The file name must be *YourName_Worksheet3.pdf*

- 1) What does logarithmic time complexity mean? Can you provide an example of the algorithms that come under this category?

Put your solution here:

- 2) A linear function takes one of the forms below. Choose the correct answer and justify your selection. let us know which type of code fragments have linear time complexity. Can you give an example?

- a) $f(n) = an + b$
- b) $f(n) = an^2 + bn + c$
- c) $f(n) = a \log_2(n)$
- d) $f(n) = a + b$

Put your solution here:

-
- 3) What is the Big-O Notation of picking a sequence of n balls from a bag of n different colored balls (the ball is not put back into the bag after being picked)?

- a) $O(n)$
- b) $O(n!)$
- c) $O(\log n)$
- d) $O(n^2)$
- e) $O(1)$

Put your solution here:

Justification:

- 4) Write the increasing order of asymptotic complexity of functions f1, f2, f3, and f4. Justify your answer.

$$f1: 2^n \log(n)$$

$$f2: n^3 + 4^n$$

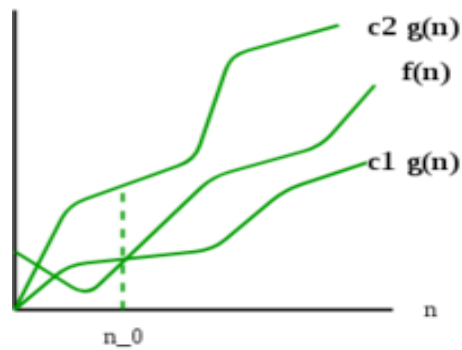
$$f3: \log(n!) + n^{1/2}$$

$$f4: n^{\log n} + 2^n$$

Put your solution here:

Justify your answer:

5) For the picture below, choose the most suitable option and justify your answer.



- a. big theta notation
- b. big oh notation
- c. big omega notation
- d. none of the above

Put your solution here:

Justification:

6) What does it mean when we say, "Algorithm A is asymptotically more efficient than B." Choose one option from below.

- a) A will be a better choice for all inputs.
- b) A will be a better choice for all inputs except possibly small inputs.
- c) A will be a better choice for all inputs except possibly large inputs.
- d) B will be a better choice for small inputs

Put your solution here:

7) Find the slowest performance time complexity from below and justify your answer.

- a) $O(n)$
- b) $O(n^2)$
- c) $O(n!)$
- d) $O(2^n)$

Put your solution here:

Justification:

8) What will be the time complexity of the following code? justify your answer?

```
value = 0
n = int(input("Enter a value: "))
for i in range(n):
    for j in range(i):
        value=value+1
```

Put your solution here:

Justification:

9) What is the running time complexity of the following code? justify your answer?

```
value = 0
val=0
n=int(input("Enter a value: "))
for i in range(n):
    for j in range(i):
        value=value+1
for i in range(n):
    for j in range (100):
        val += 1
```

Put your solution here:

Justification:

10) What is the time complexity of the below program? justify your answer.

```
n=int(input("Enter a value: "))
for i in range(1, n):
    for j in range(i,n,i):
        print("hello")
```

Put your solution here:

Justification:

11) What will be the time complexity of the following function? justify your answer.

```
def f(n):  
    while n!= 0:  
        n /= 2  
    n=int(input("Enter a value: "))  
    f(n)
```

Put your solution here:

12) What will be the time complexity of the following function? justify your answer.

```
def find_subsets(arr, n):  
    for i in range(0, 2**n):  
        subset = []  
        for j in range(0, n):  
            if (i & (1 << j)) > 0:  
                subset.append(arr[j])  
        print(subset)
```

Put your solution here:

Justification:

13) show that:

- a. $f(n) = 2n + 5$ is $O(n)$.
- b. $f(n) = 2n^2 + 3$ is $\Omega(n^2)$.
- c. $f(n) = 2n^3 - 7n + 1$ is $\Omega(n^3)$

Put your solution here:

14) Which of the following is false? Justify your selected answer.

- a. $100 n \log n = O(n \log n / 100)$
- b. $\sqrt{\log n} = O(\log \log n)$
- c. If $0 < x < y$ then $n^x = O(n^y)$
- d. $2^n \neq O(n^k)$

Put your solution here:

Justification:

15) Order the following functions by asymptotic growth rate.

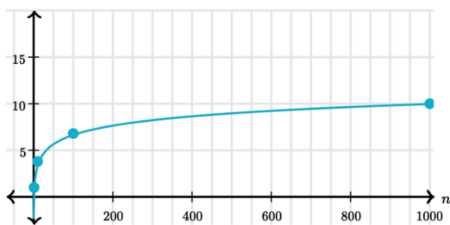
$4n \log n + 2n$, 2^{10} , $2^{\log n}$, $4n$, 2^n , $n^2 + 10n$, n^3 , $n \log n$

Put your solution here:

16) For the functions, $\log_2 n^{(\log_2 17)}$ and $\log_2 17^{(\log_2 n)}$ what is the asymptotic relationship between these functions? Write all the possible relationships for them.

Put your solution here:

17) What order of complexity does this graph represent?



Put your solution here:

18) Imagine a program A running with time complexity $\Theta(f(n))$, taking t seconds for an input of size k . What would your estimation be for the execution time for an input of size $2k$ for the following functions: n , n^2 , n^3 , 2^n , $\log_2 n$. Is this growth ratio constant for any k , or is it changing?

Put your solution here:

- 19) Consider two programs implementing algorithms A and B , trying to solve the same problem for an input of size n . They measured the execution times for test cases of different sizes and tabulated them as follows. Which program is more efficient? Why?

Algorithm	$n = 100$	$n = 200$	$n = 300$	$n = 400$	$n = 500$
A	0.003s	0.36s	0.36s	0.24s	0.053s
B	0.040s	0.160s	0.360s	0.640s	1.000s

Put your solution here:

Justification:

-
- 20) Provide True or False for the following statements:

- Algorithm A and B have a worst-case running time of $O(n)$ and $O(\log n)$, respectively. Therefore, algorithm B always runs faster than algorithm A .
- $f(n) = O(g(n))$ if there exist positive constants n_0 and c such that $f(n) \leq cg(n)$ for all $n \geq n_0$.
- O notation denotes the worst-case running time complexity with a lower bound.
- $f(n) = \Theta(g(n))$ if there exist positive constants n_0 , c_1 and c_2 such that $c_1g(n) \leq f(n) \leq c_2g(n)$ for all $n \geq n_0$.
- If for an algorithm time complexity is given by $O(n)$ then complexity of it is linear and for $O(1)$ it is said to be constant.

Put your solution here:

-
-
-
-
-